

Experiment–5

Dynamic Routing Using RIP

1. Aim / Overview of the Practical

To configure dynamic routing using Routing Information Protocol (RIP) in Cisco Packet Tracer and verify communication between devices belonging to different networks through multiple routers. The experiment also demonstrates how routers exchange routing information and update their routing tables without requiring every route to be entered manually.

2. Software / Tools Required

- Cisco Packet Tracer
- Windows Operating System

3. Theory

A) Routing

Routing is the process of determining an appropriate path for forwarding data packets from a source network to a destination network. When two devices belong to different networks, routers examine their routing tables and determine the next router or interface through which the packet should be forwarded.

B) Dynamic Routing

Dynamic routing allows routers to automatically learn and maintain routes by exchanging routing information with neighbouring routers. Unlike static routing, the administrator does not need to manually configure every destination route. When changes occur in the network, routing information can be updated automatically through the routing protocol.

Advantages of Dynamic Routing

Advantage	Description
Automatic route discovery	Routers can learn about available networks automatically.
Less manual configuration	Administrators do not have to enter every route individually.
Adaptability	Routes can be updated when changes occur in the network.
Suitable for larger networks	Managing routes becomes easier as the network grows.

Common Dynamic Routing Protocols

Protocol	Full Form
RIP	Routing Information Protocol
OSPF	Open Shortest Path First
EIGRP	Enhanced Interior Gateway Routing Protocol
BGP	Border Gateway Protocol

C) RIP – Routing Information Protocol

RIP is a distance-vector routing protocol that enables routers to exchange information about reachable networks with directly connected neighbouring routers. It provides a simple mechanism for automatically maintaining routing information when manually updating routes becomes difficult.

Working of RIP

RIP uses hop count as its routing metric. A hop represents a router that a packet needs to pass through while travelling towards its destination. For example, in the path R1 → R2 → R3 → Network X, the path has a hop count of 2 according to the routing example used in this experiment.

RIP Hop Count

Hop Count	Meaning
1	Destination network is one router away.
2	Destination network is two routers away.
5	Destination network is five routers away.
15	Maximum reachable distance using RIP.
16	Destination is considered unreachable.

Example RIP Routing Table

Destination Network	Next Hop	Hop Count
192.16.1.0	Directly Connected	0
192.16.2.0	10.0.11.2	1
192.16.3.0	10.0.11.2	2

Static Routing vs Dynamic Routing

Feature	Static Routing	Dynamic Routing
Configuration	Routes are entered manually.	Routes are learned automatically.
Route updates	Administrator must modify routes.	Routing protocol updates routes.
Administration	Becomes difficult for larger networks.	Easier to maintain as networks expand.
Best suited for	Small and stable networks.	Medium and large networks.
CPU usage	Relatively low.	Higher due to routing calculations.
Memory usage	Lower.	Higher because learned routes are stored.
Bandwidth	No routing updates are exchanged.	Bandwidth is used for routing advertisements.
Adaptability	Limited when topology changes.	Can respond to topology changes.

4. Steps for Experiment / Practical

A) Dynamic Routing with 6 PCs, 3 Routers and 3 Switches

1. Open Cisco Packet Tracer and create a topology containing 3 routers, 3 switches, and 6 PCs.
2. Connect the PCs to the switches and connect the switches to their respective routers.
3. Connect the three routers with appropriate links so that the networks can communicate.
4. Configure suitable IP addresses on all PCs and router interfaces.
5. Configure the appropriate default gateway on every PC.
6. Enable RIP Version 2 on each router.
7. Advertise the directly connected networks using the RIP configuration commands.
8. Verify the routing information using the command: show ip route
9. Check whether the routers have learned the remote networks through RIP.
10. Use the ping command between PCs belonging to different LANs to verify end-to-end connectivity.

Fig. 5.1 - RIP Dynamic Routing: 3 Routers, 3 Switches, 6 PCs

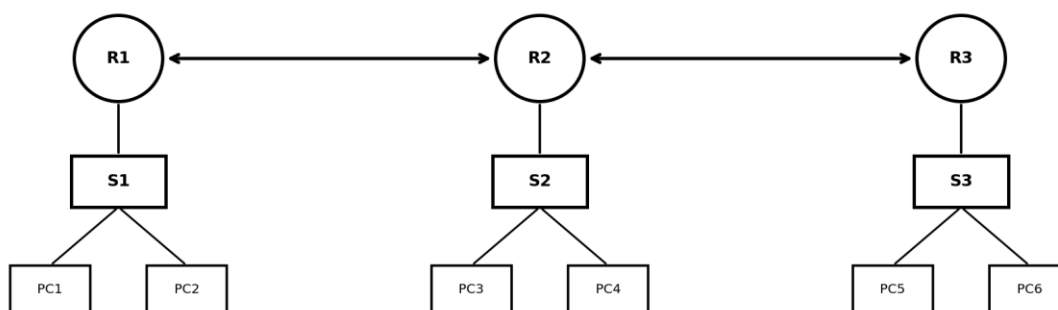


Fig. 5.1 – RIP Dynamic Routing: 3 Routers, 3 Switches and 6 PCs

B) Dynamic Routing with 8 PCs, 4 Routers and 4 Switches

11. Start Cisco Packet Tracer and place 4 routers, 4 switches, and 8 PCs in the workspace.
12. Establish the required connections between PCs, switches, and routers.
13. Assign valid IP addresses to all PCs and router interfaces.
14. Configure the correct default gateway for every PC.
15. Configure RIP Version 2 on all four routers.
16. Advertise the networks directly connected to each router.
17. Display the routing table on each router using: show ip route
18. Confirm that the remote networks have been learned through RIP.
19. Perform ping tests between PCs located on different LANs.
20. Verify that successful replies are received, confirming inter-network connectivity.

Fig. 5.2 - RIP Dynamic Routing: 4 Routers, 4 Switches, 8 PCs

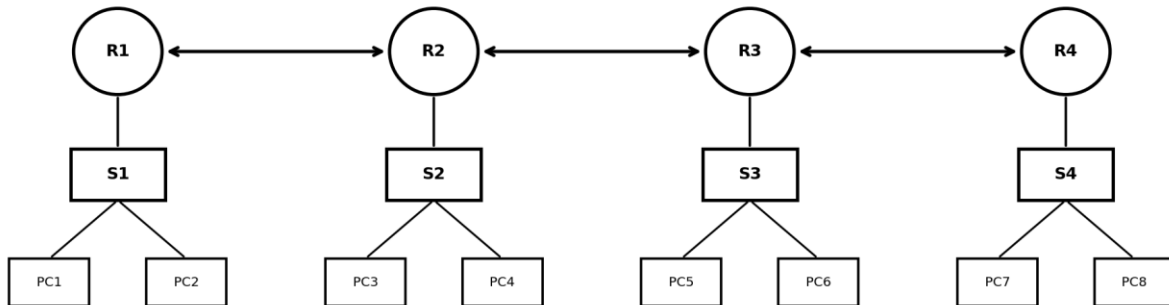


Fig. 5.2 – RIP Dynamic Routing: 4 Routers, 4 Switches and 8 PCs

5. Sample RIP Configuration

Router1 – Part A

```
Router1(config)# router rip
Router1(config-router)# version 2
Router1(config-router)# network 172.16.1.0
Router1(config-router)# network 10.0.12.0
Router1(config-router)# no auto-summary
```

6. Output

The configured topologies should demonstrate successful communication between PCs located on different LANs after RIP exchanges the required routing information.

7. Result

Dynamic routing using RIP was successfully configured in Cisco Packet Tracer for both the 3-router and 4-router network topologies. The routers exchanged routing information and automatically learned the networks that were not directly connected to them. The routing tables were verified using `show ip route`, and successful ping responses confirmed connectivity between PCs located on different networks.

8. Learning Outcomes

- Understood the purpose and importance of routing in interconnected networks.
- Identified the fundamental difference between static and dynamic routing.
- Learned the working principle of RIP as a distance-vector routing protocol.
- Understood the use of hop count as the metric in RIP.
- Configured RIP on multiple routers using Cisco Packet Tracer.
- Observed automatic exchange and learning of routing information between routers.
- Practiced verifying learned routes using the show ip route command.
- Verified connectivity between different networks using the ping command.